

Orlando Reyes

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About Me

Electronics engineering student focused on embedded systems and low-level software development. Hands-on experience with bare-metal STM32/ARM firmware, ESP32, FPGA/VHDL digital design, and Docker-based infrastructure. Experienced across the full stack, from register-level peripheral drivers to Linux and Python automation tooling.

 Get Updated CV (PDF) | Updated: September 5, 2026

Embedded & Software Projects

SCADA System for Industrial PLC

Designed and implemented a low-cost SCADA system integrating a Linux-based industrial PLC with Node-RED for real-time HMI monitoring and secure remote access via a WireGuard-based mesh VPN.

RISC-V Processor Core

Designed and implemented a custom 32-bit RISC-V processor core in VHDL for FPGA logic synthesis and functional hardware execution.

IEEE-754 Floating-Point Calculator

Developed a 32-bit floating-point arithmetic unit in VHDL and ARM Assembly featuring UART-based dynamic I/O communication.

STM32 Bare-Metal Drivers

Wrote register-level SPI drivers for ILI9341 and ST7735R TFT displays with XPT2046 resistive touch controller integration directly from datasheets.

Custom PCB & Power Hardware Design

Designed minimum-system MCU development boards in KiCad and developed power electronics topologies with custom board layouts.

FPV Flight Controller Board Debugging

Assembled an FPV quadcopter running Betaflight on an STM32F103 flight controller; resolved hardware issues on non-standard clone boards.

Self-Hosted Homelab Infrastructure

Architected and managed a self-hosted Linux NAS containerizing microservices like Home Assistant, local storage, and media streaming via Docker.

Workflow Automation & DevTools

Built Python and Playwright web scrapers/automation tools utilizing Chrome DevTools Protocol (CDP) for event management systems.

Education

Engineer in Electronics

Universidad Autónoma de Aguascalientes

08/2023 – Present (Expected 2027)

Vice Chair and founding member of the IEEE Student Branch at UAA. Experienced in hardware documentation and technical publishing using $\text{\LaTeX}$.

Technician in Mechatronics

CBTis No. 168

08/2019 – 07/2022

Hands-on training in diagnostics, maintenance of mechatronic equipment, AutoCAD drafting, and CNC programming.

Professional Experience

Database & Systems Administrator

Academia Yamaha AGS

05/2024 – 06/2026

- Standardized student databases and automated reporting workflows using custom Python scripts and VBA for Excel.
- Managed SAT-compliant electronic invoicing and inventory tracking software tools.

Team Leader IS-Warehouse

Sensata Technologies

01/2023 – 03/2024

- Built automated reporting models in Excel for real-time production tracking and schedule monitoring.
- Coordinated material logistics and stock availability based on high-volume production demand.

Technical Skills

Programming: C • C++ • VHDL • ARM Assembly • Python • VBA

Embedded Systems: STM32 (Cortex-M) • ESP32/ESP8266 • PIC • AVR

Protocols: UART/USART • SPI • I²C • CAN

Tools & OS: KiCad • Git • Docker • Linux • STM32CubeIDE • PlatformIO

Languages

English: B1 / Intermediate (Professional Reading & Writing)

Soft Skills

Team Leadership • Technical Writing • Project Coordination • Troubleshooting